# Probability before Kolmogorov

Interpretations · Worked student edition

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# What does a probability mean?

In a hypothetical agent benchmark, success was recorded on 90 of 100 tasks. What does the statement “success probability is 0.9” assert, and what does it assume?

The observed fraction is 0.9 on this benchmark. Treating it as a probability for future tasks requires a model connecting the benchmark tasks to the tasks of interest. The answer also depends on what probability is taken to mean: long-run frequency, coherent degree of belief, or a relation between evidence and a proposition. The axioms specify valid calculations; by themselves they do not select one interpretation.

# The calculus and its interpretation

Separate two questions: which probability calculations are valid, and what do the numbers represent?

Kolmogorov's 1933 *Grundbegriffe der Wahrscheinlichkeitsrechnung* axiomatized the calculus. The axioms constrain a probability measure. They do not, by themselves, say whether a probability is a long-run frequency, a degree of belief, or something else.

# Laplace's classical rule

A fair die has six equally likely faces. What probability does the classical rule assign to an even face?

The classical rule is favourable cases divided by equally possible cases:

$$P(A) = \frac{\text{number of favourable cases}}{\text{number of equally possible cases}}.$$

For a fair die, three of the six faces are even, so the rule gives $3/6 = 1/2$. Hájek, §3.1, discusses Laplace's formulation in the *Philosophical Essay on Probabilities* (1814).

# The assumption behind counting

What justifies assigning the same probability to every face?

The classical rule needs an equally-likely partition. In the die example, fairness is the model assumption that gives each face probability $1/6$. Counting then follows from that assumption. It does not establish fairness.

# A circular definition

Can “equally possible” define probability without already using a probability idea?

No. To call two cases equally possible is already to assert that neither has greater probability. The classical definition explains probability using a special case of probability. Keynes (1921) called the appeal to equal possibility the principle of indifference.

# One experiment, two partitions

Toss a fair coin twice. What is the probability of at least one head? First list the ordered outcomes.

The ordered sample space is $\{HH, HT, TH, TT\}$. Under independent fair tosses, each outcome has probability $1/4$. The event is $\{HH, HT, TH\}$, so its probability is $3/4$.

# The same event with three cases

A three-case account lists: a head on the first toss; a tail then a head; two tails. Are these cases equally likely?

They form a disjoint, exhaustive partition: $\{HH, HT\}$, $\{TH\}$ and $\{TT\}$. Their probabilities are $1/2$, $1/4$ and $1/4$. Only the first two cases include at least one head, so the probability is $1/2 + 1/4 = 3/4$. Counting the cases as equal would give $2/3$, which is incorrect for this partition.

# Partition dependence

What failed in the three-case calculation?

The list itself is valid: exactly one case occurs. The error is assigning equal weights without justification. The same experiment admits different exhaustive partitions, and their parts need not have the same probabilities.

# When the counting rule is valid

State a condition under which $P(A) = |A|/|\Omega|$ is justified for finite $\Omega$.

If the model assigns equal probability to each of the finitely many outcomes, then each has probability $1/|\Omega|$. Finite additivity gives

$$P(A) = \sum_{\omega \in A} P(\{\omega\}) = \frac{|A|}{|\Omega|}.$$

Thus counting is a consequence of the uniform finite model, not a definition that selects that model.

# A measured fraction is not a probability

In $n$ observed trials, the event $A$ occurred $N_n(A)$ times. What does $N_n(A)/n$ describe?

It is the observed relative frequency:

$$f_n(A) = \frac{N_n(A)}{n}.$$

It is a statistic computed from a finite record. A probability belongs to a model. The two quantities are related in repeated-trial settings, but they are not the same definition.

# The frequency interpretation

How does the frequency account define the probability of $A$?

It identifies probability with the limiting relative frequency of $A$ in an indefinitely long sequence of repetitions. Venn's *The Logic of Chance* (1866) and von Mises's work (1919; 1928) developed frequency accounts. Von Mises formulated them using infinite sequences called collectives.

# What is a collective?

Why did von Mises impose a condition on selected subsequences?

A collective is an infinite sequence in which relative frequencies of attributes converge and the limits remain unchanged under admissible selection rules that do not use the selected outcomes. If arbitrary selection were allowed, one could select subsequences based on outcomes and alter their frequencies. The admissible rules therefore need restriction; Church formalized place selection using recursive rules (see Hájek, §3.4).

# Limits of the frequency account

Can a finite run establish that the limiting relative frequency exists? What about a one-time event?

No finite record establishes an infinite limit. The frequency account postulates the limit rather than deriving it. It also requires an event to belong to a repeatable sequence, so it does not directly assign probability to a single non-repeatable event.

# A theorem or a definition?

Under the frequency interpretation, is convergence a theorem? Under the axioms, what is its status?

Under the frequency definition, convergence is true by construction. Under the axioms, convergence is a theorem under stated hypotheses. For independent, identically distributed Bernoulli trials with success probability $p$, the strong law gives convergence of the observed success frequency to $p$ with probability one. A theorem makes its hypotheses explicit.

# Probability as degree of belief

How might a probability describe uncertainty about an event that will not be repeated?

Subjective accounts interpret probability as a degree of belief. Ramsey (1926) and de Finetti (1931, 1937) connected a person's probability for $A$ to betting quotients they would accept. This approach can address one-off events as well as repeated trials.

# Coherence constrains beliefs

What condition prevents a system of betting probabilities from guaranteeing a loss?

Coherence requires that the assigned betting quotients do not permit a set of bets that loses in every outcome. This requirement forces finite addition rules. The calculus is recovered from a consistency condition on the degrees of belief.

# Finite and countable additivity

Does coherence alone justify countable additivity?

De Finetti argued that this reasoning justifies finite additivity, not countable additivity. That is a mathematical difference from Kolmogorov's third axiom. Several standard limit theorems fail without countable additivity.

# Probability as a relation to evidence

Can probability be understood as a relation between evidence and a proposition, rather than a personal belief?

Logical accounts treat probability as partial entailment: a relation between evidence and a proposition that is not defined by an individual's opinion. Keynes (1921), Jeffreys (1939) and Carnap (1950) developed versions of this programme. It seeks a canonical assignment when evidence is incomplete.

# The principle of indifference returns

How does the logical account face the difficulty of assigning probabilities without enough evidence?

It may appeal to a canonical assignment or the principle of indifference. But indifference depends on how possibilities are partitioned. The two-toss example showed that exhaustive cases need not be equally probable. The logical account therefore faces the same partition-dependence as the classical rule.

# Fiducial probability

What was Fisher's fiducial argument intended to obtain?

Fisher (1930) proposed deriving probability statements about an unknown parameter from its sampling distribution, without assuming a prior distribution. The aim was to obtain interval statements in applied statistics without the subjective element of a prior.

# What fiducial probability is—and is not

Is fiducial probability an alternative interpretation of the probability calculus?

It concerns inferential statements about unknown parameters; it is not an account of what probability means generally. It is therefore not an alternative to the axioms in the same sense as frequency or degree-of-belief accounts. Fisher's 1930 proposal is nearly contemporary with Kolmogorov's 1933 axiomatization. Parameter-distribution methods need their own assumptions; a sampling distribution alone does not automatically supply a unique probability law for an unknown parameter. See Reid, slides 7–8 and 16.

# Bertrand: a question without a sampling rule

If someone says “choose a chord at random,” is the probability model specified?

No. The phrase does not identify a probability measure on the chord space. Bertrand’s paradox shows that different defensible sampling mechanisms give different probabilities for the same geometric event. Specify the mechanism—for example, what quantity is uniform—and the model is well posed. This diagnosis needs no new mathematics.

# What Bertrand does not explain

Does specifying a sampling rule provide a framework for proving results about infinite sequences or processes?

No. Bertrand illustrates the need to specify the model, but it does not provide the mathematical language for infinite sequences, continuous-time processes, or conditioning on null events. Those require probability measures on suitable spaces and events.

# Measure theory preceded the axioms

Was measure theory invented to resolve Bertrand's paradox?

No. Measure and integration supply a way to assign mass to sets and calculate probabilities on infinite outcome spaces. Measure theory supplied tools for formulating and proving probability results beyond finite counting.

# Infinite sequences need measurable events

To say that relative frequencies converge with probability one, what set must be an event?

The set of infinite outcome sequences on which the frequencies converge must belong to the event σ-algebra. Countable additivity and closure under countable unions support such statements. Finite additivity alone is insufficient for the standard limit theory.

# Processes on function spaces

What kind of probability space can represent a random continuous path?

A process such as Brownian motion can be represented by a probability measure on a space of continuous functions. Kolmogorov's extension theorem gives conditions under which consistent finite-dimensional distributions determine a measure on an infinite product space.

# Conditioning on a probability-zero event

Why does $P(A \mid B) = P(A \cap B)/P(B)$ not define conditioning when $P(B) = 0$?

The denominator is zero. In continuous models, exact-value events commonly have probability zero, so the elementary ratio cannot define conditional probabilities there. Kolmogorov's conditional expectation relative to a σ-algebra supplies a more general object, defined up to events of probability zero. Specifying a value on an individual null event requires additional structure; the elementary ratio alone is insufficient.

# A shared mathematical framework

What do finite audits, continuous measurements and random paths need in common?

Each needs an outcome space, measurable events and a probability measure. Their specific spaces differ, but common axioms let us state and prove rules that apply across these models.

# What the axioms settle

Do the axioms determine which probability interpretation is correct?

No. The axioms fix rules of calculation. Interpretations differ over what probabilities assert and how a particular measure is chosen. The sample space, event collection and probability measure are parts of the model. Bertrand illustrates why they must be specified; measure theory provides a language for specifying them and proving limit theorems.

# Reading guide

Which arguments should be revisited in the main notes, and which belong to the Bertrand supplement?

Read Section 2 of the main notes alongside this deck. The two-toss partition calculation is worked within this deck and in supplement §1. Main notes §3 poses Bertrand's question; supplement §3 develops the sampling models and common chord space. This deck is a historical and conceptual companion; it does not replace the worked probability problems.

# Sources and further reading

Where can the interpretations and formal definitions be checked?

Alan Hájek, Interpretations of Probability, §§1, 3.1–3.4: calculus, classical, evidential, subjective and frequency accounts.

Nancy Reid, Distributions for Parameters, slides 7–8 and 16: Fisher's fiducial proposal and its inferential role.

Krishna Jagannathan, Probability Foundations for Electrical Engineers, Chapters 4 and 11: spaces, measurable functions and laws.